

CE/CZ4055: CPS Security

Tutorial 1

- (1) Charlie Chaplin Video: Is the eating machine a cyber physical system? If so, why is it not functioning properly? If not, why it is not a cyber physical system and what is missing to ensure that this is a cyber physical system.

Mealy: Output = $f(\text{State}, \text{input})$, Moore: Output = $f(\text{State})$ 2a, States: detected first 1, haven't detect any 1s. input, the bits, output, if state = detected first 1, flip the bits, if not don't flip

(2)

- (a) Construct a Mealy machine that calculates the 2's complement of a given input binary sequence. Assume that the bits are input to the state machine from the LSB to the MSB.

Initial state -> detect a -> next state -> detect b ... until abab then output 1, anything else go back to initial state

- (b) Construct a Moore machine that calculates the number of occurrences of the word "abab" in any input string over a,b,c.

- (c) For the Moore machine of 2(b), construct an equivalent Mealy machine.

- (3) What are the Threats (T), Vulnerabilities (V) and Risks (R) for an autonomous vehicle?